CPSC-406 Report

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Abstract

Contents

1 Introduction

2 Week by Week

2.1 Week 1

Exercise 1

Consider the following DFAs $\mathcal{A}_1$ and $\mathcal{A}_2$. Complete the following table with all 8 possible strings of length 3 over $\{a, b\}$ (columns: Accepted by $\mathcal{A}_1$? and Accepted by $\mathcal{A}_2$?).

Answer to Exercise 1.1. Here is my completed table (words of length 3 over $\{a, b\}$):

word	Accepted by $\mathcal{A}_1$?	Accepted by $\mathcal{A}_2$?
aaa	No	Yes
aab	Yes	No
aba	No	No
abb	No	No
baa	No	Yes
bab	No	No
bba	No	No
bbb	No	No

Exercise 1.2

Describe the languages $L(\mathcal{A}_k)$ accepted by $\mathcal{A}_k$ for $k = 1, 2$.

Answer to Exercise 1.2. For $k = 1$, $L(\mathcal{A}_1)$ is just the single word aab. For $k = 2$, $L(\mathcal{A}_2)$ is all words over $\{a, b\}$ that end in *at least two a's in a row* (so the last two symbols are aa).

Exercise 2 (Designing DFAs)

Design DFAs whose accepted languages are given as follows:

1. All the words that end with ab
2. All the words that contain aba
3. All the words that contain an odd number of a's and an odd number of b's
4. All the words that contain an even number of a's and an odd number of b's
5. All the words such that any three consecutive characters contain at least one a
6. All the words that contain bbb

What do you notice when comparing the various automata?

Answers for Exercise 2.

(1) Words that end with ab. I use three states that remember the last one or two symbols. The DFA is:

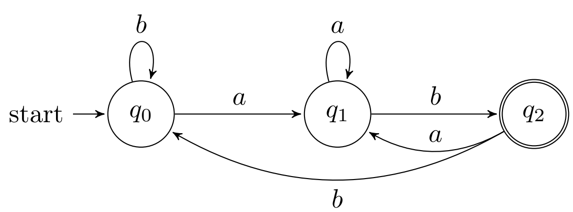


Figure 1: DFA for all words that end with ab.

(2) Words that contain aba. Here I track how much of the pattern aba I have just seen as a suffix.

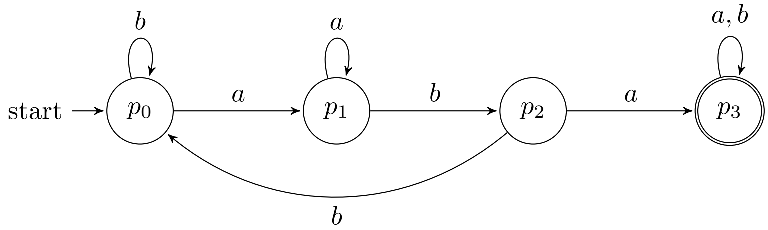


Figure 2: DFA for all words that contain aba as a substring.

(3) Odd number of a's and odd number of b's. For this and the next language I use the same 4-state parity automaton, but choose different accepting states. The four states remember the parity (even/odd) of the number of a's and b's seen so far.

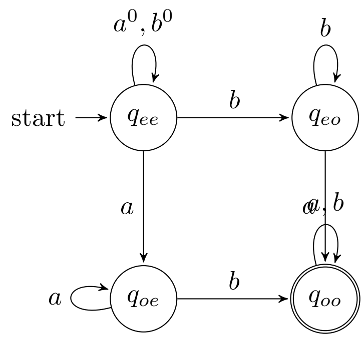


Figure 3: Parity DFA: state q_{xy} means x-parity for a's and y-parity for b's. For language (3) we accept q_{oo} (odd a's and odd b's).

(4) Even number of a's and odd number of b's. Using the same parity DFA in Figure 3, this time the accepting state is q_{eo} (even a's, odd b's) instead of q_{oo}.

(5) Any three consecutive characters contain at least one a. This is the same as saying that we never see bbb as a substring. I track runs of consecutive b's up to length 3; hitting three b's in a row moves to a dead (rejecting) state.

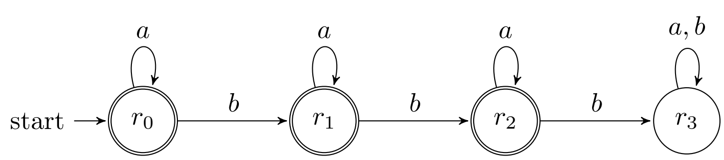


Figure 4: DFA that forbids bbb; states r_0, r_1, r_2 are accepting, r_3 is a dead state reached after three b's in a row.

(6) Words that contain bbb. I reuse the same transition structure as in Figure 4, but now r_3 is the *only* accepting state (once we have seen three consecutive b's we stay in an accepting sink).

What I notice. Several of these automata reuse the same underlying structure but just change which states are accepting (for example (3) vs. (4), and (5) vs. (6)). In other words, a lot of the "different" languages here really come from the same core DFA with a different choice of final states.

Question

How might you combine the languages $L(\mathcal{A}_1)$ and $L(\mathcal{A}_2)$ from Exercise 1 using set operations (union, intersection, complement)? Could you represent the logic with boolean logic exclusively?

2.2 Week 2: Operations on Automata

Product automata

Exercise 1 (Product automata)

Consider the following automata $\mathcal{A}^{(1)}$ and $\mathcal{A}^{(2)}$.

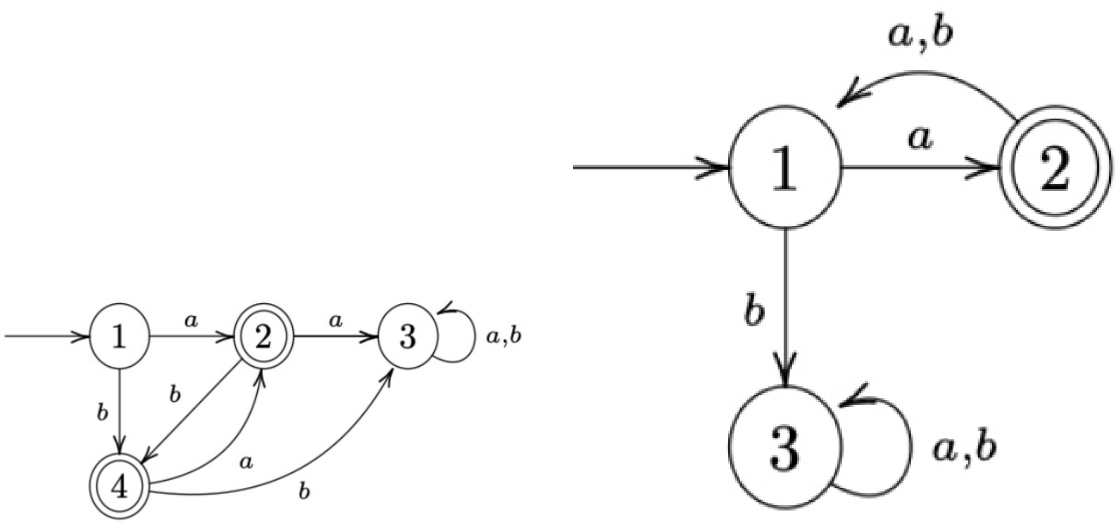


Figure 5: $\mathcal{A}^{(1)}$ (left) and $\mathcal{A}^{(2)}$ (right).

1. **Describe $L(\mathcal{A}^{(1)})$ and $L(\mathcal{A}^{(2)})$.**

   Answer:

   - $L(\mathcal{A}^{(1)})$: Words include a, b, ba, bab, $baba$, ab, aba, $abab$, etc. I can see that each character must alternate based on its successive character.
   - $L(\mathcal{A}^{(2)})$: Words include a, axa (where x doesn't matter). I see the word must start with a, and must add to its word length in increments of 2, every other character being a.

2. **Construct the intersection automaton $\mathcal{A}$ for $\mathcal{A}^{(1)}$ and $\mathcal{A}^{(2)}$ by drawing its transition diagram (omit unreachable states).**

   Answer: Below is the intersection automaton diagram.

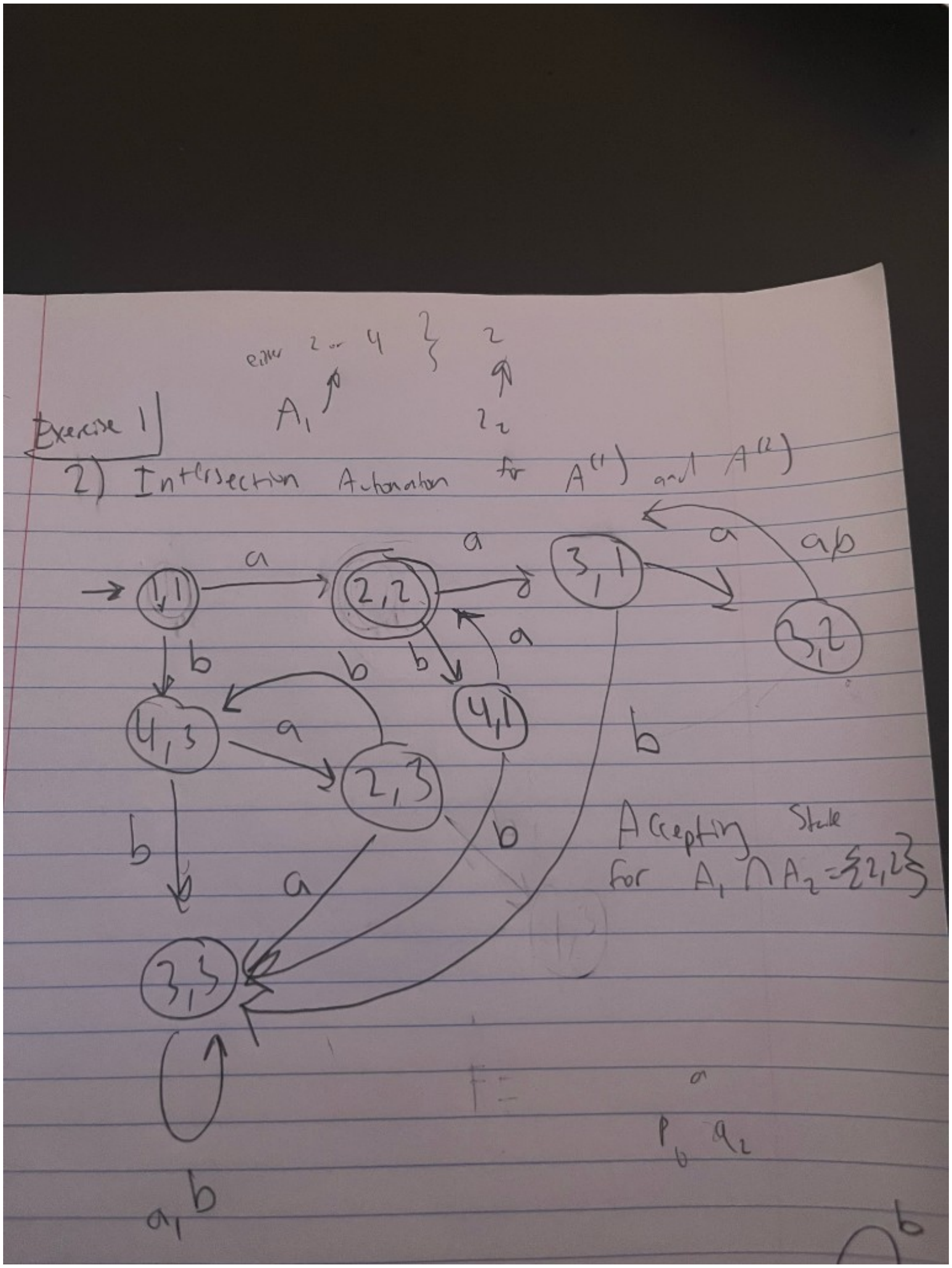


Figure 6: Intersection automaton $\mathcal{A}$ for $\mathcal{A}^{(1)}$ and $\mathcal{A}^{(2)}$ (omit unreachable states).

Note: $\{2, 2\}$ as accepting state for intersection product automaton.

3. **Argue that** $L(\mathcal{A}) = L(\mathcal{A}^{(1)}) \cap L(\mathcal{A}^{(2)})$**.**

Answer: Since $L(\mathcal{A}^{(1)})$ includes words such as a, b, ba, bab, $baba$ and ab, aba, $abab$, and $L(\mathcal{A}^{(2)})$ includes words that (1) must start with a—so we can disregard the words that start with b in $L(\mathcal{A}^{(1)})$, leaving us with a, ab, aba, etc.—and (2) the requirement for $L(\mathcal{A}^{(2)})$ is to be in increments of two, we are left with patterns: a, aba, $ababa$, etc. Thus we obtain the automaton of both intersection.

4. **How can you change $\mathcal{A}$ to get an automaton $\mathcal{A}'$ with $L(\mathcal{A}') = L(\mathcal{A}^{(1)}) \cup L(\mathcal{A}^{(2)})$?**

   Answer: In the product automaton we would add the accepting states to not be exclusive to both, but inclusive of either one of the subset DFA's accepting state.

Exercise 2 (More automata)

Consider the following automata $\mathcal{B}^{(1)}$ and $\mathcal{B}^{(2)}$.

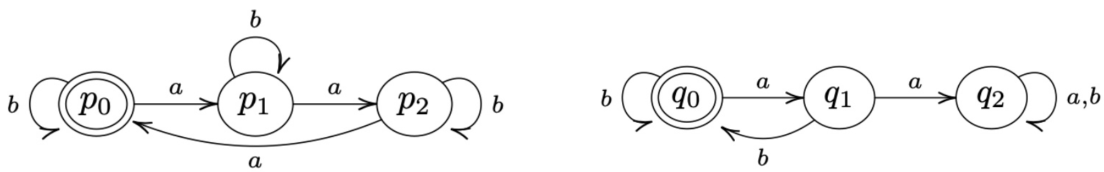


Figure 7: $\mathcal{B}^{(1)}$ (left) and $\mathcal{B}^{(2)}$ (right).

(Recall: the complement of a language $L \subseteq \Sigma^*$ is $\overline{L} := \Sigma^* \setminus L$.)

1. **Describe $L(\mathcal{B}^{(1)})$ and $L(\mathcal{B}^{(2)})$.**

   Answer:

Figure 8: $L(\mathcal{B}^{(1)})$ rules: include 3 a's at a time, any amount of b's. $L(\mathcal{B}^{(2)})$ rules: b must follow an a, any amount of b's. DFA diagram for language 2.

2. **Construct the intersection automaton $\mathcal{B}$ for $\mathcal{B}^{(1)}$ and $\mathcal{B}^{(2)}$ (omit unreachable states).**

   Answer: Below is the intersection automaton diagram.

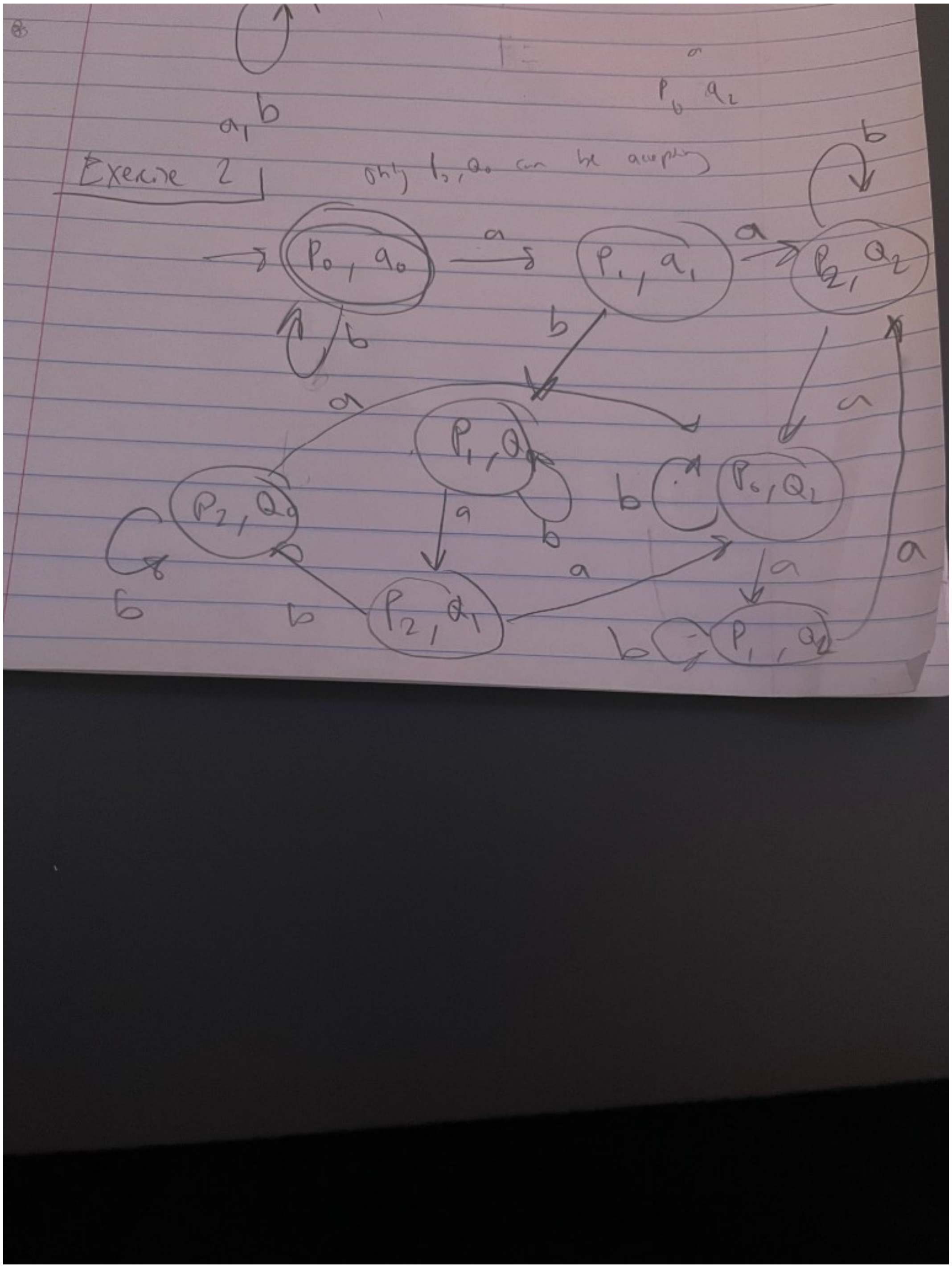


Figure 9: Intersection automaton $\mathcal{B}$ for $\mathcal{B}^{(1)}$ and $\mathcal{B}^{(2)}$ (omit unreachable states).

Note: (P_0, Q_0) as accepting state.

3. **Argue that** $L(\mathcal{B}) = L(\mathcal{B}^{(1)}) \cap L(\mathcal{B}^{(2)})$**.**

Answer: Both automata have the rule that any amount of b's is allowed. But we must combine the rules regarding a's. $L(\mathcal{B}^{(1)})$ requires 3 a's to be included at a time per word, while $\mathcal{B}^{(2)}$ requires the rule that a must be followed by a b. This gives us the conclusion that words that do include an a must come in pairs of 3 a's with b following the last a.

4. **How can you change $\mathcal{B}$ to get an automaton $\mathcal{B}'$ with $L(\mathcal{B}') = L(\mathcal{B}^{(1)}) \cup \overline{L(\mathcal{B}^{(2)})}$?**

   Answer: The Union is equivalent to its intersection, no possible change.

Exercise 3

Let $\mathcal{A}$ be a DFA and q a state of $\mathcal{A}$ such that from q, every input symbol a leaves us in q (i.e. $\delta(q, a) = q$ for all a). Show by induction on the length of the input that for any input string w, we still end up in q (i.e. $\delta^*(q, w) = q$).

Answer: We show it by induction on the length of w. (Here $\delta^*(q, w)$ is just the state we end up in after starting at q and reading the whole string w.)

When the input has length 0, the string is empty. So we don't read anything and we stay at q. So $\delta^*(q, w) = q$ in that case.

Now suppose we've already shown that for every string of length n, we end up back at q. Take a string w of length $n + 1$. We can write it as $w = va$ (some string v of length n plus one symbol a). After reading v we're in $\delta^*(q, v)$, and by our assumption that's q. Then we read a, and we're in $\delta(q, a)$, which is q by the given rule. So after reading w we're still in q, i.e. $\delta^*(q, w) = q$.

So by induction, for all input strings w we have $\delta^*(q, w) = q$.

Question

If you build the product automaton for $\mathcal{B}^{(1)}$ and $\mathcal{B}^{(2)}$ but change the accepting set to get the *union* $L(\mathcal{B}^{(1)}) \cup L(\mathcal{B}^{(2)})$ instead of the intersection, how many states must be accepting?

2.3 Week 3

Homework 1 (NFA)

1. Describe the language accepted by $\mathcal{A}$ using a regular expression.
2. Specify the NFA $\mathcal{A}$ in the form $(Q, \Sigma, \delta, q_0, F)$.
3. Find all paths in $\mathcal{A}$ for the word $w = abab$; represent in a diagram.
4. Construct the determinization $\mathcal{A}^D$ of $\mathcal{A}$ using the power set construction. Create both a transition table and a graphical depiction of $\mathcal{A}^D$.

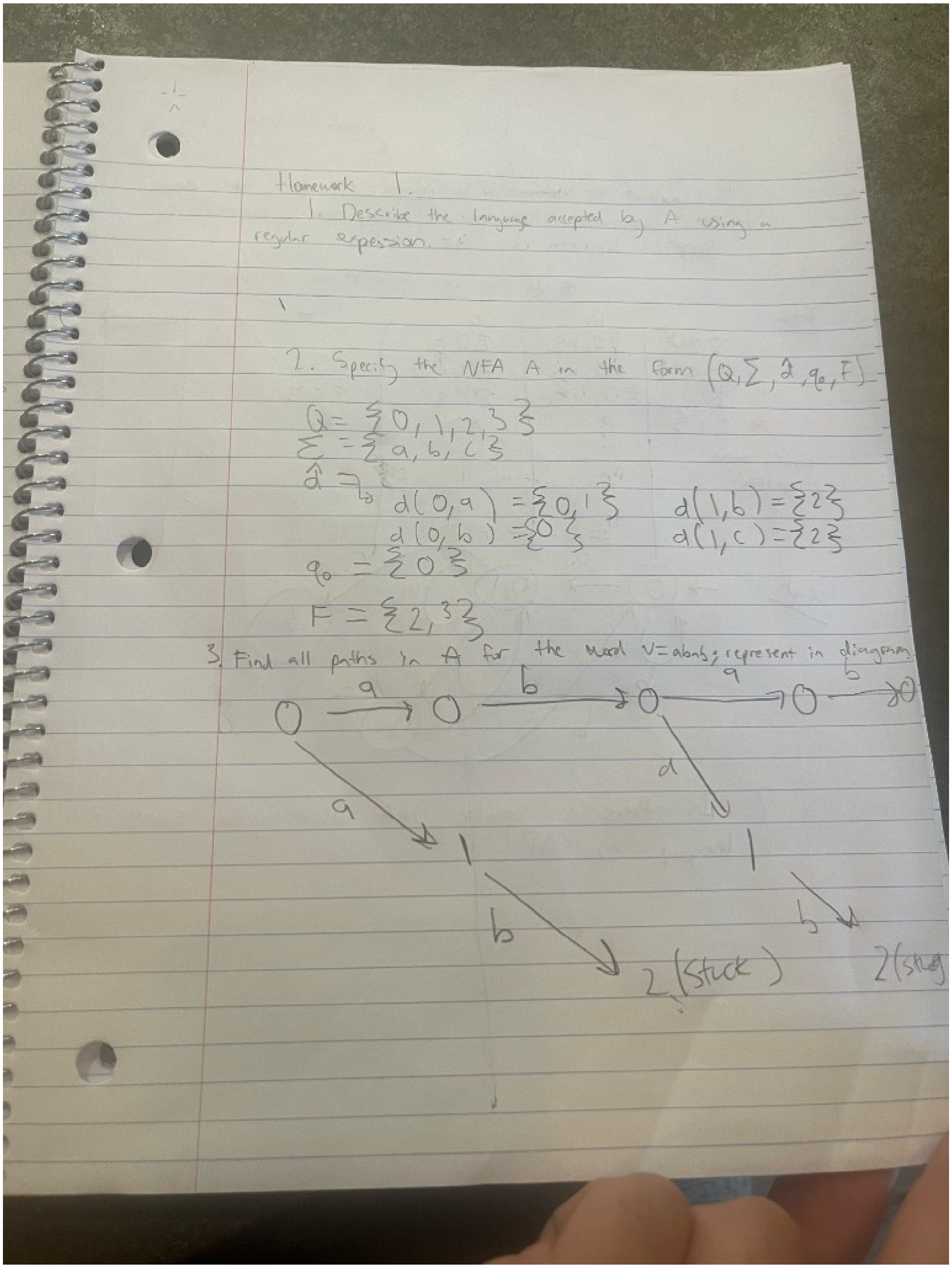
Answer (NFA specification). Here is how I write the NFA $\mathcal{A}$ formally:

$$Q = \{0, 1, 2, 3\}, \qquad \Sigma = \{a, b, c\}, \qquad q_0 = 0, \qquad F = \{2, 3\}.$$

For the transition function $\delta : Q \times \Sigma \to \mathcal{P}(Q)$ I use

$$\begin{aligned} &\delta(0, a) = \{0, 1\}, \quad &&\delta(0, b) = \{0\}, \quad &&\delta(0, c) = \emptyset, \\ &\delta(1, b) = \{2\}, \quad &&\delta(1, c) = \{2\}, \end{aligned}$$

and any transition not listed goes to the empty set (i.e. there is no move).



Homework 1.

1. Describe the language accepted by A using a regular expression.

2. Specify the NFA A in the form $(Q, \Sigma, \delta, q_0, F)$

$Q = \{0, 1, 2, 3\}$

$\Sigma = \{a, b, c\}$

$\delta \rightarrow$ $d(0,a) = \{0,1\}$ $d(1,b) = \{2\}$

$d(0,b) = \{0\}$ $d(1,c) = \{2\}$

$q_0 = \{0\}$

$F = \{2,3\}$

3. Find all paths in A for the word $v = abab$; represent in diagram

Figure 10: Homework 1: NFA specification ($Q = \{0, 1, 2, 3\}$, $\Sigma = \{a, b, c\}$, $q_0 = 0$, $F = \{2, 3\}$) and all paths for $w = abab$.

Transition table for $\mathcal{A}^D$ over $\{a,b,c\}$. From my construction I focus on the following subset-states:

$$\{0\},\ \{0,1\},\ \{0,2\},\ \{3\}.$$

The transitions on a, b, and c are:

State S	on input a	on input b	on input c
$\{0\}$	$\{0,1\}$	$\{0\}$	$\emptyset$
$\{0,1\}$	$\{0,1\}$	$\{0,2\}$	$\{3\}$
$\{0,2\}$	$\{0,1\}$	$\{0\}$	$\emptyset$
$\{3\}$	$\emptyset$	$\emptyset$	$\emptyset$

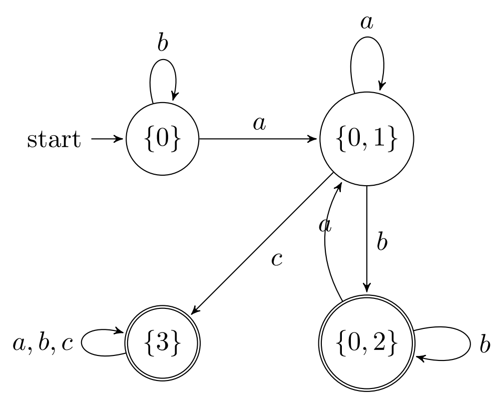


Figure 11: Determinized DFA $\mathcal{A}^D$ for Homework 2 over alphabet $\{a,b,c\}$. Accepting states are the subsets that contain an original accepting state ($\{0,2\}$ and $\{3\}$).

Homework 2 (DFA and minimization)

Determine $\mathcal{A}^D$ of $\mathcal{A}$ using the power set construction, now with alphabet $\{0,1\}$. Create both a transition table and a graphical depiction of $\mathcal{A}^D$. Then decide whether there can be a smaller DFA accepting the same language.

Transition table over $\{0,1\}$. From the NFA given in the homework, I use the following subset-states:

$$q_0,\ \{q_0,q_1\},\ \{q_0,q_1,q_2\},\ \{q_0,q_3\},\ \{q_0,q_1,q_3\},\ \{q_0,q_1,q_2,q_3\}.$$

The determinized DFA $\mathcal{A}^D$ then has this transition table:

State S	on input 0	on input 1
q_0	q_0	$\{q_0,q_1\}$
$\{q_0,q_1\}$	q_0	$\{q_0,q_1,q_2\}$
$\{q_0,q_1,q_2\}$	$\{q_0,q_3\}$	$\{q_0,q_1,q_2\}$
$\{q_0,q_3\}$	$\{q_0,q_3\}$	$\{q_0,q_1,q_3\}$
$\{q_0,q_1,q_3\}$	$\{q_0,q_3\}$	$\{q_0,q_1,q_2,q_3\}$
$\{q_0,q_1,q_2,q_3\}$	$\{q_0,q_3\}$	$\{q_0,q_1,q_2,q_3\}$

Any subset that contains the original accepting state(s) is accepting in $\mathcal{A}^D$.

Smaller DFA? Yes. In this table the last two rows (for $\{q_0,q_1,q_2,q_3\}$ and for $\{q_0,q_3\}$) have exactly the same outgoing transitions and the same accepting/non-accepting status. That means these two states are equivalent and can be merged into a single state, giving a smaller DFA that still recognizes the same language.

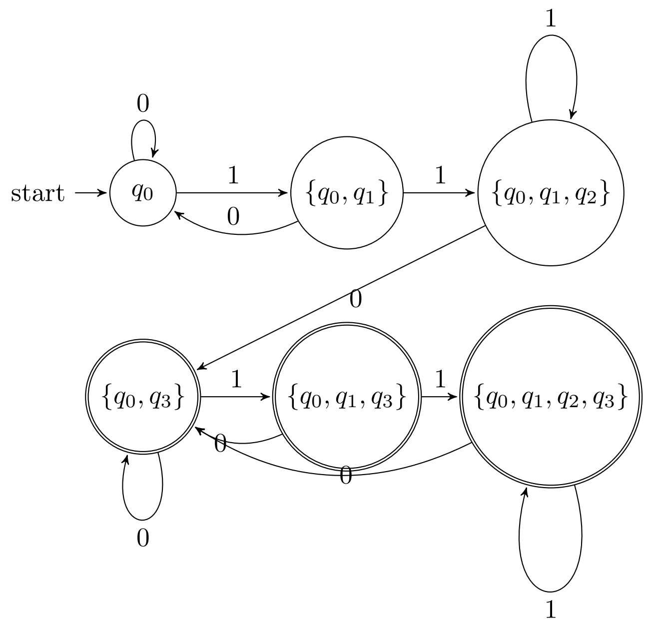


Figure 12: Determinized DFA $\mathcal{A}^D$ for Homework 2 over alphabet $\{0, 1\}$. Accepting states are the subsets that contain the original accepting state q_3.

Question

After building $\mathcal{A}^D$ from an NFA $\mathcal{A}$ using the power set construction, when can we get a smaller DFA that still accepts the same language?

2.4 Week 4

Consider the following DFA $\mathcal{A}$: there is a state p (initial and final) with transition a to itself and transition b to state q; state q has transition b to itself and transition a to p.

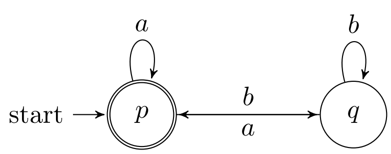


Figure 13: DFA $\mathcal{A}$ for Week 4: p is initial and final; $p \xrightarrow{a} p$, $p \xrightarrow{b} q$, $q \xrightarrow{b} q$, $q \xrightarrow{a} p$.

Question. For $p = 1$ and $q = 2$, compute via Kleene's algorithm all regular expressions $R_{ij}^{(k)}$ and from this a resulting regular expression R describing $L(\mathcal{A})$, i.e., such that $L(R) = L(\mathcal{A})$.

Answer. I label the states so that state 1 is p and state 2 is q. In Kleene's algorithm, $R_{ij}^{(k)}$ denotes the set of all strings that label paths from i to j with intermediate states (if any) in $\{1, \ldots, k\}$. The recurrence is

$$R_{ij}^{(k)} = R_{ij}^{(k-1)} \cup R_{ik}^{(k-1)} \big(R_{kk}^{(k-1)}\big)^* R_{kj}^{(k-1)}.$$

I write union as + in expressions below.

Base case $k = 0$ (no intermediate states; only direct edges and ε on same state):

$$R_{11}^{(0)} = \varepsilon + a, \qquad R_{12}^{(0)} = b, \qquad R_{21}^{(0)} = a, \qquad R_{22}^{(0)} = \varepsilon + b.$$

Inductive step $k = 1$ (intermediate state 1 allowed). Using $(R_{11}^{(0)})^* = (\varepsilon + a)^* = a^*$:

$$\begin{aligned}
R_{11}^{(1)} &= R_{11}^{(0)} + R_{11}^{(0)}(R_{11}^{(0)})^* R_{11}^{(0)} = (\varepsilon + a) + (\varepsilon + a)a^*(\varepsilon + a) = a^*;\\
R_{12}^{(1)} &= R_{12}^{(0)} + R_{11}^{(0)}(R_{11}^{(0)})^* R_{12}^{(0)} = b + (\varepsilon + a)a^* b = a^* b;\\
R_{21}^{(1)} &= R_{21}^{(0)} + R_{21}^{(0)}(R_{11}^{(0)})^* R_{11}^{(0)} = a + a\,a^*(\varepsilon + a) = a\,a^*;\\
R_{22}^{(1)} &= R_{22}^{(0)} + R_{21}^{(0)}(R_{11}^{(0)})^* R_{12}^{(0)} = (\varepsilon + b) + a\,a^* b = (\varepsilon + b) + a^+ b.
\end{aligned}$$

Inductive step $k = 2$ (all states allowed). Set $C = R_{22}^{(1)} = (\varepsilon + b) + a^+ b = b + a^+ b$ (since $\varepsilon + b$ gives ε and b). Then $(R_{22}^{(1)})^* = (b + a^+ b)^*$. Only state 1 is accepting, so $L(\mathcal{A})$ is the set of strings that label paths from 1 to 1:

$$R_{11}^{(2)} = R_{11}^{(1)} + R_{12}^{(1)} \left(R_{22}^{(1)}\right)^* R_{21}^{(1)} = a^* + a^* b\,(b + a^+ b)^*\, a^+.$$

Thus a regular expression R with $L(R) = L(\mathcal{A})$ is

$$\boxed{R \;=\; a^* + a^* b\,(b + a^+ b)^*\, a^+}.$$

(Here $a^+ = aa^*$; one can also write $a^* b\,(b \cup a^+ b)^*\, a^+$ if union is denoted $\cup$.)

Question

If we changed the DFA so that both p and q were accepting states, how would the regular expression R for $L(\mathcal{A})$ change? Would it simplify to a shorter expression?

2.5 Week 5

Exercise (Pumping lemma and non-regular languages)

Show that the following languages are not regular using the pumping lemma:

$$L_1 = \{a^n b^m \in \{a, b\}^* \mid n, m \in \mathbb{N},\ n > m \geq 0\},$$

$$\begin{aligned}
L_2 &= \{a^{n^2} \in \{a\}^* \mid n \in \mathbb{N},\ n \geq 1\},\\
L_3 &= \{a^p \in \{a\}^* \mid p \in \mathbb{N},\ p \text{ prime}\},\\
L_4 &= \{a^k b^m a^{k+m} \mid k, m \geq 0\}.
\end{aligned}$$

1. Showing L_1 is not regular. I assume L_1 is regular and take a pumping constant p from the lemma. I pick the string

$$w = a^{p+1} b^p \in L_1,$$

since here $n = p+1$ and $m = p$ so indeed $n > m$. By the pumping lemma I can write $w = xyz$ with $|xy| \leq p$, $|y| \geq 1$, and for all $i \geq 0$ the pumped string $xy^i z$ should still be in L_1.
Because $|xy| \leq p$ and w starts with only a's, the substring y must consist entirely of a's. So say $y = a^t$ with $t \geq 1$. If I pump down with $i = 0$, I get the string

$$xy^0 z = xz = a^{p+1-t} b^p,$$

so the number of a's in this string is $p + 1 - t$, and the number of b's is still p. Since $t \geq 1$, I now have $p + 1 - t \leq p$, so $n \leq m$ for this pumped string. That means $xy^0 z \notin L_1$, contradicting the pumping lemma. So my assumption that L_1 was regular must be wrong; L_1 is not regular.

2. Showing L_2 is not regular. Again I assume L_2 is regular and let p be the pumping length. I choose

$$w = a^{p^2} \in L_2,$$

since p^2 is a perfect square. The lemma gives me a split $w = xyz$ with $|xy| \le p$, $|y| \ge 1$, and $xy^iz \in L_2$ for all $i \ge 0$. Because the whole word is just a's and $|xy| \le p$, I know that $y = a^t$ for some $1 \le t \le p$.
Now I pump once with $i = 2$. Then

$$xy^2z = a^{p^2+t}.$$

If this were in L_2, I would need $p^2 + t = n^2$ for some integer n. But $1 \le t \le p$, so $p^2 < p^2 + t < (p+1)^2 = p^2 + 2p + 1$. There is no perfect square strictly between p^2 and $(p+1)^2$, so $p^2 + t$ cannot be a square. Hence $xy^2z \notin L_2$, contradicting the pumping lemma. So L_2 is not regular.

3. Showing L_3 is not regular. I assume L_3 is regular and take its pumping length p. Choose $w = a^q \in L_3$ where q is any prime number with $q \ge p$ (there are infinitely many primes, so I can definitely pick one large enough). Then $w = xyz$ with $|xy| \le p$, $|y| \ge 1$, and $xy^iz \in L_3$ for all $i \ge 0$. As before, $y = a^t$ where $1 \le t \le p$.
The length of w is q, and the length of xy^iz is $q + (i-1)t$. I look at $i = q + 1$ to pump a bunch of copies of y:

$$|xy^{q+1}z| = q + qt = q(1 + t).$$

This number is clearly a composite (product of q and $1 + t$, both > 1), so it is not prime. That means $xy^{q+1}z \notin L_3$, contradicting the pumping lemma. Therefore L_3 is not regular.

4. Showing L_4 is not regular. Assume L_4 is regular and let p be its pumping length. I pick

$$w = a^p b^p a^{2p} \in L_4,$$

since here $k = p$ and $m = p$, so the last block is $a^{k+m} = a^{2p}$. The pumping lemma says $w = xyz$ with $|xy| \le p$, $|y| \ge 1$, and $xy^iz \in L_4$ for all $i \ge 0$. Because $|xy| \le p$ and the first p symbols are a's, the piece y sits entirely in the first block of a's. So $y = a^t$ for some $1 \le t \le p$.
If I pump with $i = 2$, the new word is

$$xy^2z = a^{p+t} b^p a^{2p}.$$

In this string, the leftmost block of a's has length $k' = p + t$, the b-block has length $m' = p$, but the rightmost block is still a^{2p}, i.e. length $2p$. For this to be in L_4 I would need the last block to have length $k' + m' = (p + t) + p = 2p + t$. That would mean I need $2p = 2p + t$, which is impossible since $t \ge 1$. So $xy^2z \notin L_4$, contradicting the pumping lemma. Hence L_4 is not regular.

Question

In these pumping-lemma proofs I always picked one very specific string w in each language. How much freedom do I actually have in choosing w? Could I have picked a different family of strings (for example, $a^{p+2}b^{p+1}$ for L_1) and still made the same argument work just as well?

3 Synthesis

4 Evidence of Participation

5 Conclusion

References

[BLA] Author, Title, Publisher, Year.